

Sharp Continuous-Order Comparison for Modified-Bessel Ratios: a Riccati Proof, the Optimal Real-Order Domain, and Lean Verification

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Abstract

For $x > 0$ put $\rho_\alpha(x) = I_{\alpha+1}(x)/I_\alpha(x)$, where I_α is the modified Bessel function of the first kind. We study the sharp continuous-order comparison

$$0 < \rho_\mu(x) - \rho_\nu(x) < \frac{\nu - \mu}{x}.$$

The lower inequality is the classical strict order decrease of the Bessel ratio. For the upper inequality we give a direct two-flow Riccati proof. If $h = \rho_\mu - \rho_\nu$, $b = (\nu - \mu)/x$, and $h = b$ at a contact point, then

$$h' = -\frac{(\nu - \mu)(\mu + \nu + 1)}{x^2}, \quad (h - b)' = -\frac{(\nu - \mu)(\mu + \nu)}{x^2}.$$

The second identity, not the first, is the contact orientation. It proves the bound for $\mu + \nu > 0$. The degenerate boundary $\mu + \nu = 0$ follows from the connection formula between I_{-a} , I_a , and K_a together with their Wronskian. Conversely the large- x expansion gives counterexamples whenever $\mu + \nu < 0$. Thus, on the natural positivity range $-1 < \mu < \nu$, the upper bound holds for every $x > 0$ if and only if $\mu + \nu \geq 0$.

The nonnegative-order statement itself is known: it is equivalent to Lemma 10 of Freitas–Laugesen. Their proof already contains a first-contact argument, so no novelty is claimed for that mechanism. The contributions here are the explicit sharp ratio formulation, the optimal extension/obstruction across negative orders, the correction of an earlier unit-step paper statement, and a Lean 4/Mathlib formalization of the complete nonnegative-order upper bound. The Lean artifact has no `sorry`; its axiom oracle returns exactly `propext`, `Classical.choice`, and `Quot.sound`. We retain the resulting Feynman–Hellmann application: every character sector gap in two-dimensional Wilson $U(1)$ and $SU(2)$ lattice gauge theory decreases strictly with the bare coupling.

1 Introduction

The ratios $\rho_\nu(x) = I_{\nu+1}(x)/I_\nu(x)$ of modified Bessel functions of consecutive orders appear throughout applied analysis, probability and statistical physics, and a substantial literature is devoted to bounding them by algebraic functions of ν and x [1, 2, 3, 4, 5, 6]. This note concerns the behaviour under arbitrary real order shifts of the logarithmic derivative

$$\frac{\partial}{\partial x} \log I_\nu(x) = \frac{I'_\nu(x)}{I_\nu(x)}.$$

Theorem 1 (sharp continuous-order comparison). *Let $x > 0$ and $0 \leq \mu < \nu$. Then*

$$0 < \rho_\mu(x) - \rho_\nu(x) < \frac{\nu - \mu}{x}. \quad (1)$$

Equivalently,

$$\frac{I'_\mu(x)}{I_\mu(x)} < \frac{I'_\nu(x)}{I_\nu(x)}. \quad (2)$$

The left inequality is classical; the proof below addresses the sharp right inequality directly.

Theorem 2 (optimal real-order domain). *Let $-1 < \mu < \nu$. The lower inequality in (1) holds for every $x > 0$. The upper inequality holds for every $x > 0$ if and only if*

$$\mu + \nu \geq 0.$$

If $\mu + \nu < 0$, then the upper inequality fails for all sufficiently large x .

Remark 3 (priority and corrected scope). The nonnegative-order conclusion (2) is Lemma 10 of Freitas and Laugesen [7]. A previous version of this note stated that fractional steps did not follow from its method and described their proof as a Bessel-zero argument. Both statements were wrong. Their published proof defines the logarithmic-derivative difference, uses its positive small- x sign, and excludes a first zero directly from the modified-Bessel differential equation. The two-flow calculation below is the same contact geometry written for ρ_μ and ρ_ν . We therefore claim neither the nonnegative-order theorem nor the first-contact mechanism as new. Segura’s Riccati/nullcline analysis [8] is also close in method, although its stated monotonicity results vary x at fixed order rather than compare two arbitrary orders. The genuinely new candidate isolated here is the optimal negative-order boundary in Theorem 2, together with the formal verification.

The purpose of this corrected note is threefold.

First, we expose the entire fractional-order comparison as one Riccati contact identity. This avoids Bessel zeros and high-order uniform asymptotics in the proof region $\mu + \nu > 0$.

Second, we resolve the contact-degenerate line $\mu + \nu = 0$ and show by asymptotics that crossing it reverses the large- x sign. This produces an exact global domain rather than a sufficient condition.

Third, we record a consequence in two-dimensional lattice gauge theory (we are not aware of an explicit statement in the literature, though it may well be folklore): by the Feynman–Hellmann identity — exact here, because the 2D transfer operator is diagonal in a β -independent character basis — the continuous-order comparison of Theorem 1 is precisely the strict decrease in the bare coupling β of every mass gap between character sectors, for the $U(1)$ and $SU(2)$ Wilson actions simultaneously (Section 4). We emphasize the modest scope: this is a statement about the exactly soluble two-dimensional theory; nothing here is claimed about four-dimensional gauge theory.

Section 5 presents the machine-checked Lean 4/Mathlib verification of the full nonnegative-order upper theorem (no `sorry`; standard axioms only). Section 6 reports an independent numerical audit.

2 The two-flow Riccati proof and the optimal domain

Write $\delta = \nu - \mu > 0$ and

$$h(x) = \rho_\mu(x) - \rho_\nu(x), \quad b(x) = \frac{\delta}{x}, \quad G(x) = h(x) - b(x).$$

The derivative and recurrence identities imply the Riccati equation

$$\rho'_\alpha = 1 - \rho_\alpha^2 - \frac{2\alpha + 1}{x} \rho_\alpha. \quad (3)$$

Subtracting the equations at μ and ν gives

$$h' = -(\rho_\mu + \rho_\nu)h - \frac{2\mu + 1}{x}h + \frac{2\delta}{x}\rho_\nu. \quad (4)$$

At a point where $h = b = \delta/x$, substitute $\rho_\mu = \rho_\nu + \delta/x$ into (4). Direct cancellation yields

$$h' = -\frac{\delta(\mu + \nu + 1)}{x^2}, \quad G' = h' - b' = -\frac{\delta(\mu + \nu)}{x^2}. \quad (5)$$

This distinction is essential: the first displayed derivative is the derivative of the ratio difference, whereas first-contact orientation is controlled by the second.

For $\alpha > -1$, the defining series gives

$$\rho_\alpha(x) = \frac{x}{2(\alpha + 1)} \left(1 - \frac{x^2}{4(\alpha + 1)(\alpha + 2)} + O(x^4) \right). \quad (6)$$

Consequently $h(x) > 0$ for small x , while $G(x) \rightarrow -\infty$ as $x \downarrow 0$. If $\mu + \nu > 0$ and G had a first zero c , then $G < 0$ immediately to its left but (5) would give $G'(c) < 0$. Differentiability would instead force $G > G(c) = 0$ immediately to the left, a contradiction. Hence $G < 0$ for all $x > 0$. This proves the upper half of Theorem 1 and the sufficiency assertion of Theorem 2 away from the boundary. Notice also that at any putative zero of h , (4) gives

$$h' = \frac{2\delta}{x} \rho_\nu > 0.$$

Together with the small- x sign this is the analogous first-contact orientation for the lower inequality. More generally, strict order decrease of $\alpha \mapsto \rho_\alpha(x)$ on $(-1, \infty)$ is a classical consequence of order log-concavity; see [10]. We use that standard result for the full lower statement.

It remains to handle the degenerate line $\mu + \nu = 0$. Write $\mu = -a$, $\nu = a$, where $0 < a < 1$. The connection formula [13, 10.27.4]

$$I_{-a}(x) = I_a(x) + \frac{2}{\pi} \sin(\pi a) K_a(x) \quad (7)$$

has a strictly positive coefficient of K_a . The Wronskian [13, 10.28.2] gives

$$\left(\frac{K_a}{I_a} \right)' = \frac{K'_a I_a - K_a I'_a}{I_a^2} = -\frac{1}{x I_a(x)^2} < 0. \quad (8)$$

Thus $I_a/I_{-a} = (1 + c_a K_a/I_a)^{-1}$ is strictly increasing. Equivalently, $(\log I_a)' - (\log I_{-a})' > 0$, which by $(\log I_\alpha)' = \rho_\alpha + \alpha/x$ is precisely $\rho_{-a} - \rho_a < 2a/x$. This proves strictness on the boundary.

Finally, the standard large- x expansion is

$$\rho_\alpha(x) = 1 - \frac{\alpha + \frac{1}{2}}{x} + \frac{4\alpha^2 - 1}{8x^2} + O(x^{-3}). \quad (9)$$

Hence

$$h(x) - \frac{\delta}{x} = -\frac{\delta(\mu + \nu)}{2x^2} + O(x^{-3}). \quad (10)$$

When $\mu + \nu < 0$, the right-hand side is positive for all sufficiently large x , proving necessity and completing Theorem 2.

For a concrete audit witness, at 100 decimal digits,

$$(\mu, \nu, x) = (-0.8, -0.4, 10), \quad \rho_\mu - \rho_\nu = 0.042682961675102858 \dots > 0.04 = \frac{\nu - \mu}{x}.$$

3 The unit-step Amos calibration

Throughout, $x > 0$, $\nu \geq 0$, I_ν is the modified Bessel function of the first kind, all values $I_\nu(x)$ are strictly positive, and $\rho_\nu(x) = I_{\nu+1}(x)/I_\nu(x) \in (0, 1)$. We use two classical facts, valid for all real orders (see [13, 10.29.1–10.29.2]):

$$I_{\nu-1}(x) - I_{\nu+1}(x) = \frac{2\nu}{x} I_\nu(x), \quad (11)$$

$$I'_\nu(x) = \frac{1}{2}(I_{\nu-1}(x) + I_{\nu+1}(x)). \quad (12)$$

Specializing Theorem 1 to the unit shift gives

$$\frac{I'_{\nu+1}(x)}{I_{\nu+1}(x)} - \frac{I'_\nu(x)}{I_\nu(x)} = \frac{1}{x} - (\rho_\nu(x) - \rho_{\nu+1}(x)) > 0. \quad (13)$$

The older four-step proof of this specialization is retained because it records the exact equivalence with the Amos bound.

Step 1: a closed form for the logarithmic derivative

Dividing (12) by I_ν and using (11) in the form $I_{\nu-1}/I_\nu = \rho_\nu + 2\nu/x$:

$$\frac{I'_\nu(x)}{I_\nu(x)} = \frac{1}{2} \left(\frac{I_{\nu-1}}{I_\nu} + \frac{I_{\nu+1}}{I_\nu} \right) = \frac{1}{2} \left(\rho_\nu + \frac{2\nu}{x} + \rho_\nu \right) = \rho_\nu(x) + \frac{\nu}{x}. \quad (14)$$

Step 2: reduction to a ratio inequality

By (14), for the unit step $\nu \rightarrow \nu + 1$,

$$\frac{I'_{\nu+1}}{I_{\nu+1}} - \frac{I'_\nu}{I_\nu} = \rho_{\nu+1} - \rho_\nu + \frac{1}{x},$$

which is (13). Writing the recurrence (11) at order $\nu + 1$ as $\rho_{\nu+1} = 1/\rho_\nu - 2(\nu + 1)/x$, the inequality $\rho_\nu - \rho_{\nu+1} < 1/x$ is equivalent to

$$\frac{1}{\rho_\nu(x)} - \rho_\nu(x) > \frac{2\nu + 1}{x}. \quad (15)$$

Step 3: the Amos-type bound

We use the classical upper bound (Amos [1]; proved best of its form and valid for all $\nu \geq 0$ by Segura [5, Thm. 2], after the index shift $\nu \mapsto \nu + 1$):

$$\rho_\nu(x) < U_\nu(x) := \frac{x}{a+s}, \quad a = \nu + \frac{1}{2}, \quad s = \sqrt{a^2 + x^2}. \quad (16)$$

Step 4: the exact calibration

Lemma 4 (calibration). *Let $a, x > 0$ and $s = \sqrt{a^2 + x^2}$, $U = x/(a+s)$. Then*

$$\frac{1}{U} - U = \frac{2a}{x},$$

and for every t with $0 < t < U$,

$$\frac{1}{t} - t > \frac{2a}{x}.$$

Proof. Since $(s-a)(s+a) = s^2 - a^2 = x^2$, we have $U = x/(a+s) = (s-a)/x$, hence

$$\frac{1}{U} - U = \frac{a+s}{x} - \frac{s-a}{x} = \frac{2a}{x}.$$

The map $t \mapsto 1/t - t$ is strictly decreasing on $(0, \infty)$, so $t < U$ gives $1/t - t > 1/U - U = 2a/x$. $\square$

Applying Lemma 4 with $t = \rho_\nu(x) < U_\nu(x)$ and $a = \nu + \frac{1}{2}$ yields (15) strictly, hence (13) and the unit-shift instance of (1). $\square$

Remark 5 (equivalence with the Amos bound). The chain of Step 2 consists of equivalences: since $t \mapsto 1/t - t$ is a strictly decreasing bijection of $(0, \infty)$ onto $\mathbb{R}$, and U_ν is precisely the positive root of $1/t - t = (2\nu + 1)/x$ (Lemma 4), the inequality (15) is *equivalent* to $\rho_\nu < U_\nu$. Therefore: *the unit-step order-monotonicity (13) is equivalent to the classical Amos upper bound (16)*. Any of the many proofs of the latter [1, 3, 4, 5] yields the former, and conversely the continuous monotonicity of Freitas–Laugesen [7] yields, in particular, a Bessel-zero proof of the Amos bound. The bound U_ν is also recognizable as the b_0 -type bound of [3]: the positive root of the Riccati-derived quadratic $t^2 + \frac{2\nu+1}{x}t - 1 = 0$.

4 Application: bare-coupling monotonicity of all 2D sector gaps

Consider lattice gauge theory on a two-dimensional rectangular lattice with Wilson action and compact structure group G , at bare coupling $\beta > 0$. We fix notation carefully, separating the three objects involved.

(a) *Character coefficients.* The one-plaquette Boltzmann weight expands as (see, e.g., [14, 15])

$$e^{\beta \operatorname{Re} \chi_f(U)} = \sum_r d_r c_r(\beta) \chi_r(U), \quad c_r(\beta) = \frac{1}{d_r} \int_G \overline{\chi_r(U)} e^{\beta \operatorname{Re} \chi_f(U)} dU,$$

the sum over irreducible representations r of G , $d_r = \chi_r(e)$. For the two classical cases ($n \in \mathbb{Z}_{\geq 0}$, $j \in \frac{1}{2}\mathbb{Z}_{\geq 0}$; verified independently to 30 digits in the audit of Section 6):

$$G = U(1): \quad c_n(\beta) = I_n(\beta), \quad G = SU(2): \quad c_j(\beta) = \frac{I_{2j+1}(2\beta)}{\beta},$$

the $SU(2)$ formula being the statement that the Haar class integral equals $\int_G \chi_j(U) e^{\beta \chi_f(U)} dU = (2j+1) I_{2j+1}(2\beta)/\beta$, i.e. $d_j c_j(\beta)$ with $d_j = 2j+1$ (all $SU(2)$ characters are real, so $\text{Re } \chi_f = \chi_f = 2 \cos \theta$).

(b) *Transfer eigenvalues.* In two dimensions, gauge fixing reduces the theory to a product of independent plaquette weights, and the transfer operator (one lattice row, temporal gauge) is diagonal in the character basis $\{\chi_r\}$ — the standard exact solubility of 2D lattice gauge theory [14, 11]; see [12] for a textbook treatment of the character expansion — a basis independent of β — with eigenvalue in sector r proportional to $c_r(\beta)$, the proportionality constant being a sector-independent normalization. Ratios of transfer eigenvalues between sectors are therefore exactly $c_r(\beta)/c_{r'}(\beta)$, with every normalization convention cancelling.

(c) *Sector gaps.* For two sectors with $c_r(\beta) > c_{r'}(\beta) > 0$ (in both models the coefficient is strictly decreasing in the sector label, so “heavier” sectors have smaller coefficients), the mass gap of sector r' relative to r is

$$m_{r,r'}(\beta) = \log \frac{c_r(\beta)}{c_{r'}(\beta)} > 0,$$

e.g. for $U(1)$ the fundamental string tension is $m_{0,1} = \log(I_0/I_1)$ and the $v87$ bracket instance is $m_{1,2} = \log(I_1/I_2)$. Because the eigenbasis is β -independent, $\frac{d}{d\beta} m_{r,r'}$ acts on the eigenvalues alone — the Feynman–Hellmann identity in its exact, diagonal form; no perturbation theory is involved.

Corollary 6. *For 2D Wilson lattice gauge theory:*

(i) $G = U(1)$: for all integers $0 \leq n < n'$,

$$\frac{d}{d\beta} m_{n,n'}(\beta) = \frac{d}{d\beta} \log \frac{I_n(\beta)}{I_{n'}(\beta)} = (\rho_n - \rho_{n'}) - \frac{n' - n}{\beta} < 0.$$

Every sector gap — in particular the string tension $\log(I_0/I_1)$ and the bracket instance $\log(I_1/I_2)$ — is strictly decreasing in β .

(ii) $G = SU(2)$: for all $0 \leq j < j'$ in $\frac{1}{2}\mathbb{Z}$,

$$\frac{d}{d\beta} m_{j,j'}(\beta) = 2 \left[(\log I_{2j+1})'(2\beta) - (\log I_{2j'+1})'(2\beta) \right] < 0,$$

the $1/\beta$ prefactors cancelling in the ratio $c_j/c_{j'} = I_{2j+1}(2\beta)/I_{2j'+1}(2\beta)$, which is free of representation-dimension factors.

Both assertions are direct instances of Theorem 1; the orders happen to be integer-spaced, so the older unit-step proof in Section 3 would also suffice.

Proof. (i) By (14), $\frac{d}{d\beta} \log(I_n/I_{n'}) = (\rho_n + n/\beta) - (\rho_{n'} + n'/\beta) = (\rho_n - \rho_{n'}) - (n' - n)/\beta$, and (1) gives $\rho_n - \rho_{n'} < (n' - n)/\beta$ strictly. (ii) Identical, at argument $x = 2\beta$ and orders $2j+1 < 2j'+1$; the chain rule contributes the factor 2. $\square$

Remark 7 (scope). Corollary 6 is a statement about the exactly soluble two-dimensional theory. The analogous monotonicity $m'(\beta) \leq 0$ for the four-dimensional mass gap on a weak-to-strong coupling corridor is an open problem (universally expected in lattice physics, covered by no theorem); the present corollary is the exact 2D instance of that bracket and nothing more. In the terminology of the audit-first programme to which this note belongs, the 2D instance is *signed*; the 4D bracket remains a named hypothesis.

5 The machine-checked algebraic core

The continuous-order upper theorem on $0 \leq \mu < \nu$ is formalized in the C4 module `FractionalOrder.lean`. Unlike the earlier standalone algebraic artifact, this file works on C4’s actual Gamma-series definition of I_ν , invokes the proved real-order Riccati equation and Amos bound, supplies a uniform small- x seed, and executes the inf first-contact argument. It contains no `sorry` and introduces no project axiom. The checked endpoint is:

```
theorem besselRatioReal_fractional_upper
  (mu nu : Real) (hmu : 0 <= mu) (hmunu : mu < nu)
  {x : Real} (hx : 0 < x) :
  besselRatioReal mu x - besselRatioReal nu x
    < (nu - mu) / x

theorem besselIReal_logDeriv_fractional_lt
  (mu nu : Real) (hmu : 0 <= mu) (hmunu : mu < nu)
  {x : Real} (hx : 0 < x) :
  deriv (fun y => Real.log (besselIReal mu y)) x
    < deriv (fun y => Real.log (besselIReal nu y)) x
```

The implementation exposes three audit lemmas: the derivative of the signed gap, its exact derivative at contact, and the small- x seed. The axiom oracle for all five declarations returns exactly:

```
[propext, Classical.choice, Quot.sound]
```

Verification used Lean 4.29.0-rc6 and pinned Mathlib commit

```
07642720480157414db592fa85b626dafb71355b.
```

On 4 August 2026 the module was built in a Colab Pro+ CPU/high-memory runtime: 8165 Lake jobs completed, the module’s `.olean` was materialized, and the oracle file compiled. Runtime opening and teardown were recorded; the runtime was disconnected and deleted after 18 minutes 53 seconds.

Honest scope. The machine-checked theorem covers the sharp upper inequality on the non-negative order axis. The classical lower inequality and the negative-order boundary/obstruction in Theorem 2 are proved on paper but are not represented in C4, whose real-order Bessel interface currently assumes nonnegative order. Source presence is not reported as verification: the claim above refers to the materialized Colab build and its oracle output.

6 Numerical audit

The fractional-order statement was attacked independently at 70 decimal digits on 800 deterministic adversarial points (seed 490060), including tiny order separations, values near $\mu + \nu = 0$, and arguments from the small- to large- x regimes. No failure occurred in the valid nonnegative domain; this is diagnostic, not proof. The same script verifies the explicit invalid-domain witness following (10). In addition, the older unit-step/application inequalities were checked at 50 decimal digits (mpmath), on the grid $\beta \in \{10^{-2}, \dots, 10^4\}$ (16 values, geometrically spread),

$\nu \in \{0, 1, \dots, 29\} \cup \{50, 100, 500\}$: the closed form (14) holds to $1.4 \cdot 10^{-18}$ (central finite difference at step 10^{-20}); the Amos margin $U_\nu - \rho_\nu$ is strictly positive throughout, with minimum $2.5 \cdot 10^{-9}$ attained at $(\beta, \nu) = (10^4, 0)$ (the bound is asymptotically sharp, as it must be, being extremal of its form); the quantitative slack $1/\beta - (\rho_\nu - \rho_{\nu+1})$ is strictly positive throughout, with minimum $1.5 \cdot 10^{-8}$ at $(10^4, 1)$; and $m(\beta) = \log(I_1/I_2)$ has $m'(\beta) < 0$ at every grid point, ranging from -1.96 at $\beta = \frac{1}{2}$ to $-1.9 \cdot 10^{-5}$ at $\beta = 278$. The $SU(2)$ Haar class integral $\int_G \chi_j e^{\beta \chi_f} dU = \frac{2}{\pi} \int_0^\pi \sin \theta \sin((2j+1)\theta) e^{2\beta \cos \theta} d\theta$ was verified to equal $(2j+1) I_{2j+1}(2\beta)/\beta$ to 30 digits at $j \in \{0, 1, 2\}$, $\beta \in \{0.7, 2\}$ — hence $c_j = I_{2j+1}(2\beta)/\beta$ after dividing by $d_j = 2j+1$. The verification script and the sources of this note are archived in the programme repository:

[https:](https://github.com/lluiseiriksson/THE-ERIKSSON-PROGRAMME/tree/main/papers/bessel-amos-fh)

[//github.com/lluiseiriksson/THE-ERIKSSON-PROGRAMME/tree/main/papers/bessel-amos-fh](https://github.com/lluiseiriksson/THE-ERIKSSON-PROGRAMME/tree/main/papers/bessel-amos-fh)

Version and audit note

This note is part of an audit-first research programme whose records, Lean sources and numerical suites are public at <https://github.com/lluiseiriksson/THE-ERIKSSON-PROGRAMME> (machine-checked core) and the satellite repositories linked there. The author thanks the adversarial-audit process itself. It first located [7], correcting a novelty claim, and later exposed a second error in version 6: the paper said its method did not close fractional steps and misdescribed Lemma 10 as a Bessel-zero proof. The contact calculation (5) corrects that statement. This version separates the known nonnegative-order theorem and its known first-contact route from the candidate-new optimal negative-order classification. Numerical residuals and a finite mesh are never used as proof.

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